



Introducing Data Science to Lws

DEEPAK PANDEY

“Empowering Learning With Analysis”

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Overview

This document summarizes the 10+ analytical approaches applied to the student's marks dataset (MEL VS FEL scores). Each methods addresses key questions about performance patterns, gaps, and predictive insights. In-depth analyses were done with covering all the aspect of data. Samples of Statical Learning, Deep learning and Machine learning is provided as for demo- project.

As a data Science Enthusiast, I have manipulated data's, recognized analysis pattern, preformed prediction of data and visualization of data. Data science not only involves performing analytical tasks but also encompasses a wide range of processes including data collection, observation, preprocessing, exploration, modeling, interpretation, and communication of insights. It integrates techniques from statistics, computer science, and domain expertise to uncover patterns, support decision-making, and increase efficiency across various industries. *All the analysis and visualization were done through coding only*

1.Introduction

In the field of education, assessing student performance across various subjects is essential for evaluating learning outcomes, identifying strengths and weaknesses, and guiding academic improvements. This dataset represents a collection of student scores from two different assessments: MEL (Midterm Exam Level) and FEL (Final Exam Level) across three subjects—English, Science, and Social Studies.

The analysis focuses on examining how students perform in both midterm and final evaluations, with the intent to derive insights that can help educators, administrators, and curriculum planners make data-driven decisions.

Purpose of the Analysis

The primary purpose of this analysis is to:

Evaluate student performance in MEL and FEL across three core subjects.

Calculate total scores for each subject by combining MEL and FEL scores.

Analyzing progress, variance , differences, gap across Subject (MEL VS FEL).

Compute a grand total per student to understand overall academic standing.

Identify trends and disparities in performance between subjects or students.

Support academic planning and targeted interventions where needed.

Scope of the Analysis

This analysis is limited to:

- A dataset of **25 students**.
- Performance scores from **MEL (out of 35)** and **FEL (out of 50)** for English, Science, and Social Studies.
- Derived fields such as **Total English, Total Science, Total Social, and Grand Total**, which currently have null values and will be computed as part of the data processing.

It does **not** include demographic data, attendance, or other behavioral indicators.

Data were provided as in below format:

2.Student Exam analysis

This section provides a detailed examination of student performance across three core academic subjects: English, Science, and Social Studies. The analysis draws upon two assessment phases—Midterm Exam Level (MEL) and Final Exam Level (FEL)—to evaluate consistency, progress, and subject-wise performance patterns.

2.1 Subject-Wise Performance

Each student's score is recorded separately for MEL (out of 35 marks) and FEL (out of 50 marks) for each subject. By combining these scores, we calculate the Total Score in each subject (out of 85 marks). This allows for a fair comparison across subjects and assessment phases.

2.2 Total and Grand Totals

To understand each student's overall academic performance, total scores for each subject will be computed and used to derive a Grand Total. This serves as a cumulative measure of performance and can be used for ranking or identifying high-performing and at-risk students.

2.3 Trend and Comparative Analysis

Through this analysis, we aim to:

Compare individual and group performance between MEL and FEL.

Identify subjects where students generally perform better or worse.

Observe if there is a significant improvement or decline from MEL to FEL.

Detect any outliers (students who perform exceptionally high or low).

Understand performance distribution across the class.

2.4 Insights and Observations

Based on the computed totals and visualized distributions (e.g., using bar graphs, line plots, or heatmaps), key observations will be drawn to highlight:

Subject proficiency trends.

The overall academic standing of students.

Recommendations for instructional improvement or student support.

3.Data Manipulation

Since the provided data were limited, empty or less variant as shown in below:

#	Column	Non-Null Count	Dtype
0	Student Name	25 non-null	object
1	MEL English (35)	25 non-null	int64
2	MEL Science (35)	25 non-null	int64
3	MEL Social (35)	25 non-null	float64
4	FEL English (50)	25 non-null	float64
5	FEL Science (50)	25 non-null	float64
6	FEL Social (50)	25 non-null	float64
7	Total English	0 non-null	float64
8	Total Science	0 non-null	float64
9	Total Social	0 non-null	float64
10	Grand Total	0 non-null	float64

dtypes: float64(8), int64(2), object (1)

memory usage: 2.3+ KB

we need to change or manipulate datasets for performing in-depth analysis.

Some of the manipulation is aimed to describe below:

- Since provided data were in MEL (35) and FEL (50) with different marks 35 and 50. There was difficulty giving conclusion over the visualization graph cause the MEL figure range from 0-35 and FEL 0-50 which will show always a below graph of MEL. So, at first estimation of the MEL (50 EST) were done from the same data of MEL(35) through statical and mathematical formula.
- As data of total Science, Social, English and Grand total were empty. We filled all the empty sets by adding (MEL and FEL) data set and converting to percentage for better insights while performing analysis.
- Total MEL and Total FEL were calculated from all MEL and FEL Subject for the comparison of the midterm and Final term analysis.

- Students were categorized into (Need Improvements, Average, good, excellent on the basis of Grand Total.
- From the MEL of two subject (Science, Social) prediction of Grade Total were made through machine learning and statical learning
- Variance were calculated between MEL subject and FEL subject in terms of Z - means.
- And last the performance gap between MEL and FEL were calculated through machine learning.

121]:

	Student Name	FEL English (50)	FEL Science (50)	FEL Social (50)	MEL English (50 Est)	MEL Science (50 Est)	MEL Social (50 Est)	Total English	Total Science	Total Social	Grand Total	Total MEI	Performance Category	Predicted Total	Total FEL
0	Aaradhya Agrahari	47.86	35.71	47.14	48.57	44.29	47.86	96.43	80.00	95.00	90.48	93.81	Excellent	91.011859	87.14
1	Aaravi Paudel	37.14	31.43	40.00	44.29	35.71	37.14	81.43	67.14	77.14	75.24	78.09	Good	72.756946	72.38
2	Aarush Thapa Chauhan	35.71	22.86	32.86	41.43	25.71	35.71	77.14	48.57	68.57	64.76	68.57	Average	66.513931	60.95
3	Aayushi Senchury	33.57	34.29	28.57	35.71	38.57	33.57	69.28	72.86	62.14	68.09	71.90	Average	69.136130	64.29
4	Abia Subedi	44.29	48.57	45.71	45.71	40.00	44.29	90.00	88.57	90.00	89.52	86.67	Good	84.316511	92.38

Form above figure, You can see all the data manipulation from the raw data provide and all the required data were created among them i want to highlight some columns to provide you some summary conclusion of above that -

- All the data were created from coding only and the accurate data is presented.
- The **Performance Category** classify all the student on their performance of Overall data.
- The main Highlight of above data is **Predicted Total**, as i mention above the predicted total is generated from the data of MEL Science and MEL Social only. Hence it has predicted close data to the overall **Grand Total which** was calculated from data of both MEL and FEL. This conclude that Data Science can perform data (analysis, distingue, prediction) for growth in future and current to increase growth of company.

4. DATA VISUALIZATION

Data visualization is a powerful tool for transforming raw data into meaningful insights, making complex information easier to understand and act upon. In the context of student performance analysis, effective visualizations help educators, administrators, and policymakers identify trends, strengths, weaknesses, and opportunities for improvement.

Why Data Visualization Matters

1. Simplifies Complex Data

- Exam scores, subject-wise breakdowns, and predicted vs. actual performance metrics can be overwhelming in tabular form. Visualizations like bar charts, line graphs, and heatmaps simplify comparisons and highlight key patterns.

2. Reveals Trends & Patterns

- Histograms and box plots show score distributions, helping identify whether most students are performing well or if there are significant gaps.
- Scatter plots and correlation heatmaps reveal relationships between different subjects (e.g., do students strong in Science also perform well in English?).

3. Identifies Outliers & Key Performers

- Box plots and percentile rankings help detect underperforming or exceptionally high-achieving students who may need targeted interventions or enrichment.

4. Facilitates Decision-Making

- Visual comparisons between **(FEL)** and **(MEL)** scores help assess preparation levels before final exams.
- Performance category breakdowns (e.g., "Excellent," "Good," "Needs Improvement") allow educators to allocate resources effectively.

5. Enhances Communication

- A well-designed dashboard with interactive charts (e.g., filters for subject-wise performance) makes it easier to present findings to stakeholders like teachers, parents, and school boards.

Let us start our insights conclusion through overall data. All the conclusions can't be explained in this document; we can perform board members meeting to cover all the areas of data.

Overall data insights

	FEL English (50)	FEL Science (50)	FEL Social (50)	MEL English (50 Est)	MEL Science (50 Est)	MEL Social (50 Est)	Total English	Total Science	Total Social	Grand Total	Total MEL	Total FEL	Predicted Total	Performa
count	25.000000	25.000000	25.000000	25.000000	25.000000	25.000000	25.000000	25.000000	25.000000	25.000000	25.000000	25.000000	25.000000	2
mean	38.600400	33.315200	36.171200	41.313600	33.941600	38.600400	79.914000	67.256800	74.771600	73.980800	75.904400	72.056400	73.980800	
std	6.361347	8.548968	6.957247	6.519464	7.839653	6.361347	12.068641	14.528763	12.068185	11.912539	11.854526	12.787876	11.099957	
min	27.140000	21.430000	22.860000	24.290000	7.140000	27.140000	53.580000	32.850000	52.150000	52.850000	47.610000	50.960000	46.884520	
25%	33.570000	27.140000	30.000000	37.140000	28.570000	33.570000	72.860000	57.140000	67.860000	65.710000	68.570000	64.290000	66.513931	
50%	37.860000	32.860000	34.290000	41.430000	35.710000	37.860000	79.280000	65.720000	74.280000	74.760000	75.720000	70.470000	72.756946	
75%	44.290000	40.000000	40.000000	47.140000	38.570000	44.290000	90.000000	80.000000	82.860000	83.330000	86.670000	83.810000	82.723326	
max	47.860000	48.570000	47.140000	50.000000	44.290000	47.860000	97.860000	92.860000	95.000000	94.290000	94.770000	93.810000	91.011859	1

From above figure, the conclusion was provided of overall data where.....

Mean: defines the average marks scores by the student out of 50marks. for example we can see first column of data **FEL English (50)** on average of **36** marks where obtain by students .

std: defines how data is spread form mean i.e. 6.36 percentage of data is clustering near the average marks of 36.

min: defines the minimum scores data among all the 25 student data where 27 marks is the lowest marks among all students in FEL English(50).

25%: We call it low range data i.e. In FEL English(50) students' scores below 33 marks out of 50 and needs improvement

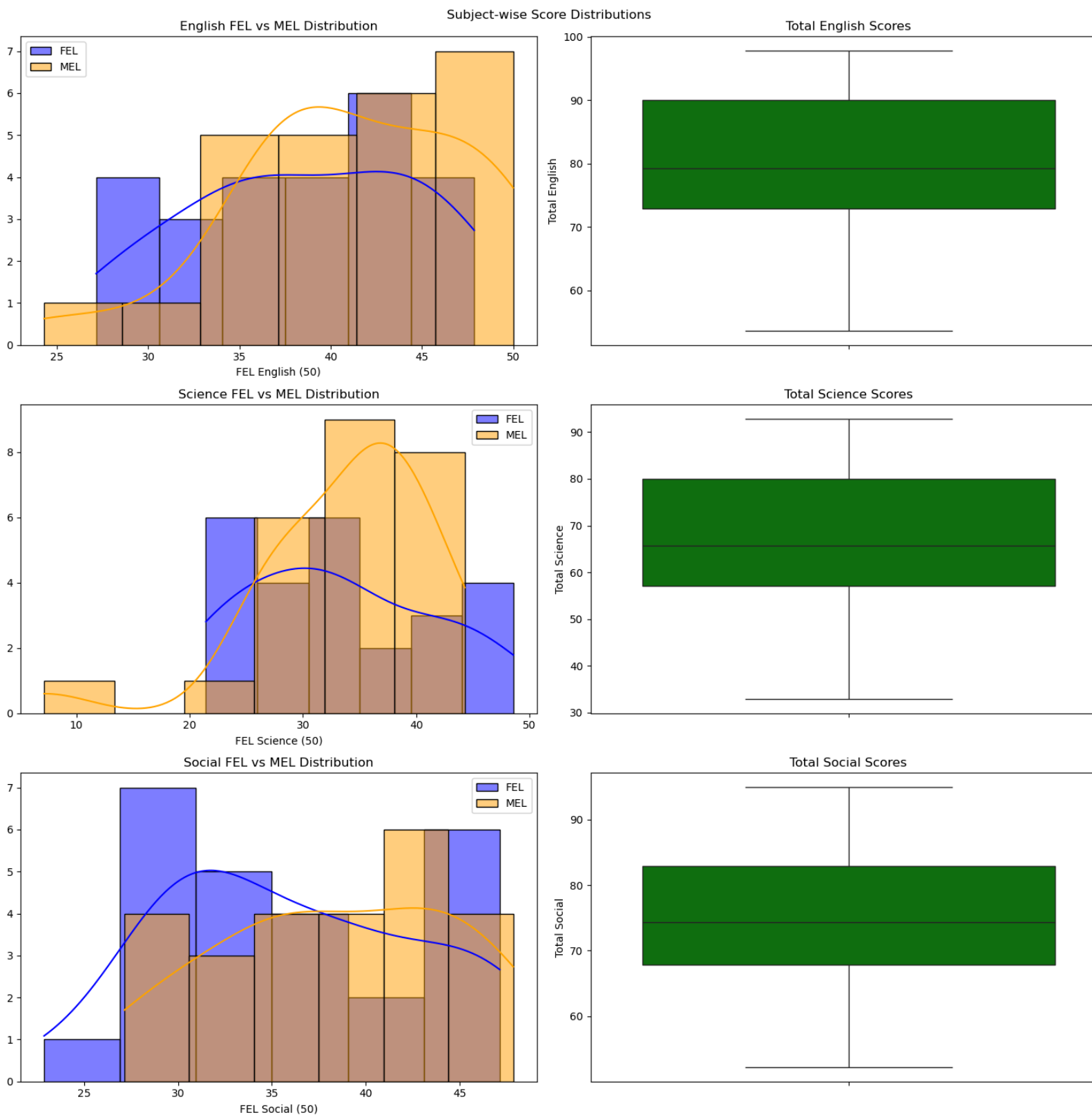
50% : We call it average or baseline data. where FEL English (50) students' scoring above / below of 37marks out of 50 are in average categories.

75%: We call it high scores i.e. In FEL English (50) students' scoring 44marks out of 50 are the top scores.

Conclusion can be given to every column as above.

Subject (MEL) vs Subject (FEL)

In this section, we have classified data for analysis of each subject Science, Social, English individually or between two datasets of MEL and FEL. Without making it longer i have visualized all the subject on one set as below:



Description and Conclusion of the above figure:

From above figure the left-hand side is the data graph of score of students in MEL and FEL whereas right -hand side figure is the data of Total scores in subjects. Each subject visualization is provided accordingly for conclusion let me explain in example:

Conclusion of Science subject:

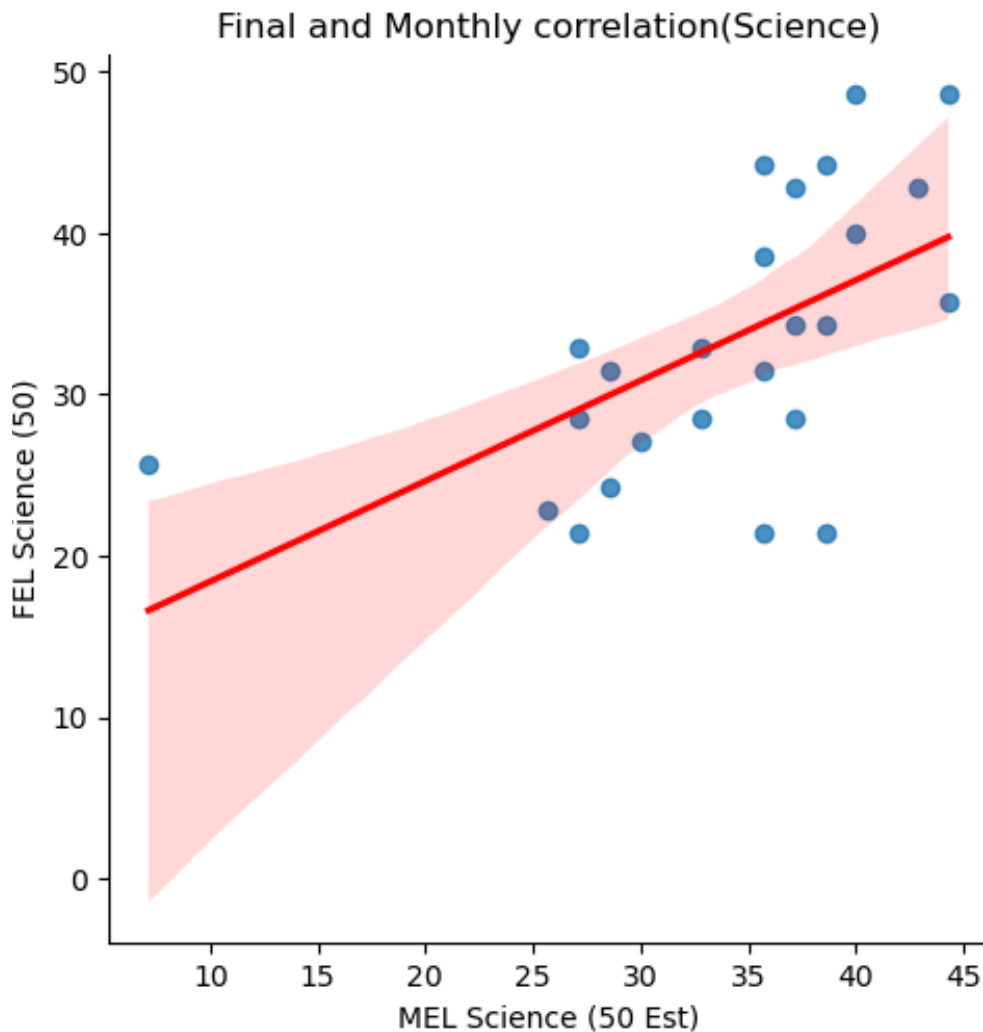
Science subject visualization data is in 2nd figure. Where the left-hand data shows marks variance of students, yellow line is of MEL and blue line is FEL. As you see in above figure the yellow line starts from 10 which says one of the students has scores less than 10 marks in MEL science whereas blue FEL line starts with 25 which says lowest scores is in range of 20 -25 .

The higher and lower lines give conclusion of most of the student's marks range. As we see on average blue and yellow lines are in range 22-45 . Where higher note of yellow line of MEL gives conclusion of most of the student marks are in this range. And blue line from range 40 -50 indicates some of the student has scores marks in this range.

On the other hand, Right- Side figure showing total science which is total of MEL and FEL gives conclusion of on average students most of the student are in range of 55-80 marks which is shown in boxlike colored structure in which central line shows the exact marks the student are range which is 65. Whereas lower side of the line from the box shows below scoring marks as students the longer the line the more students are in the range of below marks which in range 65-35 marks. However, the upper line from the box shows the higher scoring student's data in range 80 -95 marks.

same as conclusion can be generated for others subject and visualization figure.

Now lets check for the Science subject overall analysis of MEL and FEL



The above figure shows the correlation of data between MEL and FEL i.e how the students are performed in MEL and FEL . On above figure the middle line is the strong bond line and outer line are the average marks range in comparison of MEL and FEL . The dots indicate the data of students.

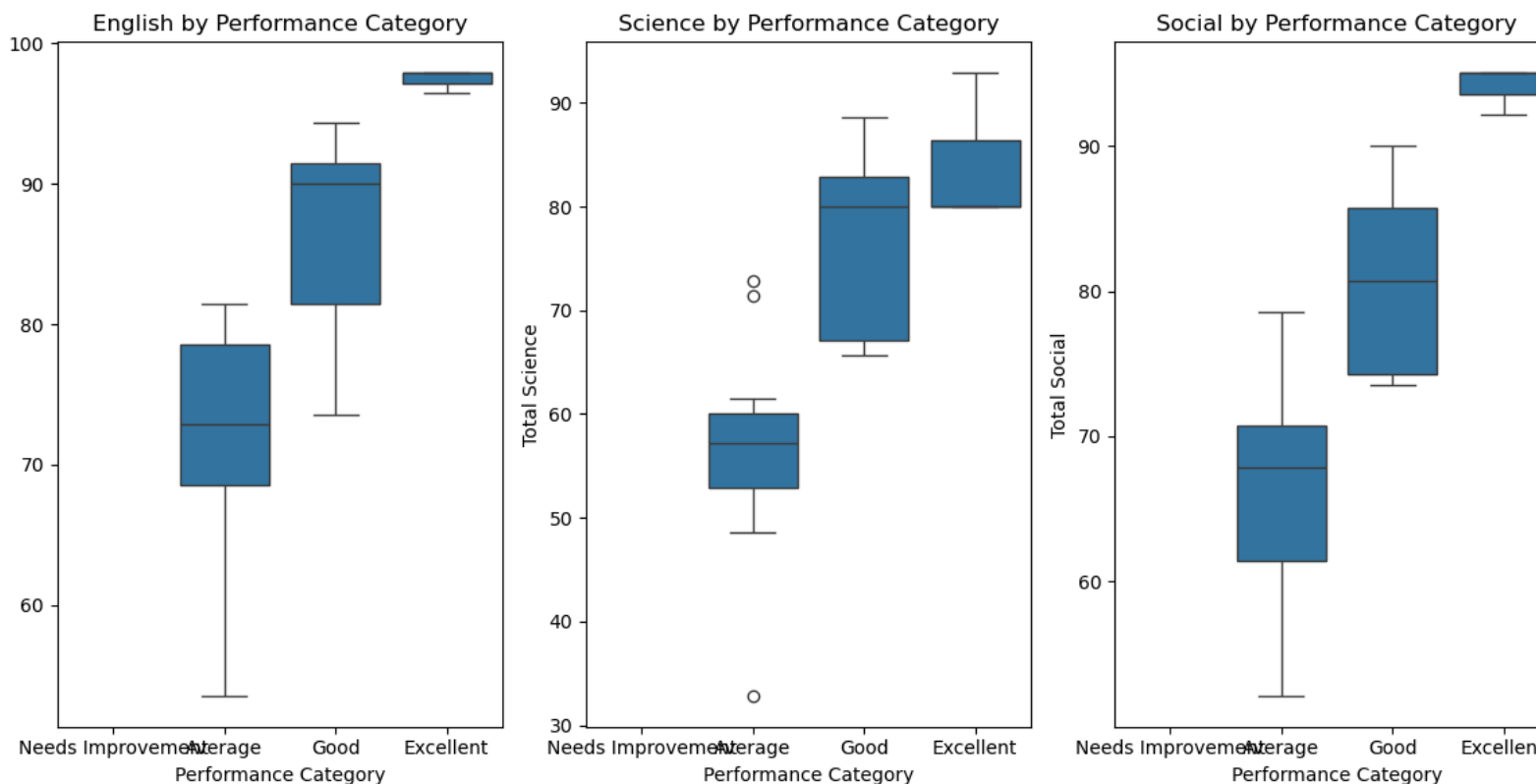
Conclusion of the above Scatter figure:

The Dots touching or closure to the red line gives conclusion of those who have scored moderate marks in MEL have also scored moderate marks in FEL. Around 4-5 students can be range from the above figure dots

Whereas dots below the line indicates the risk of student who has gradually decrease in FEL in comparison to Mel report and Upper dots indicates higher marks data.

Subject wise Performance Category:

The Subject-wise Performance Category provides a detailed breakdown of a student's academic performance across individual subjects. This categorization helps identify areas of strength, subjects needing improvement, and overall academic balance. It serves as a diagnostic tool for educators, students, and parents to support targeted learning strategies.



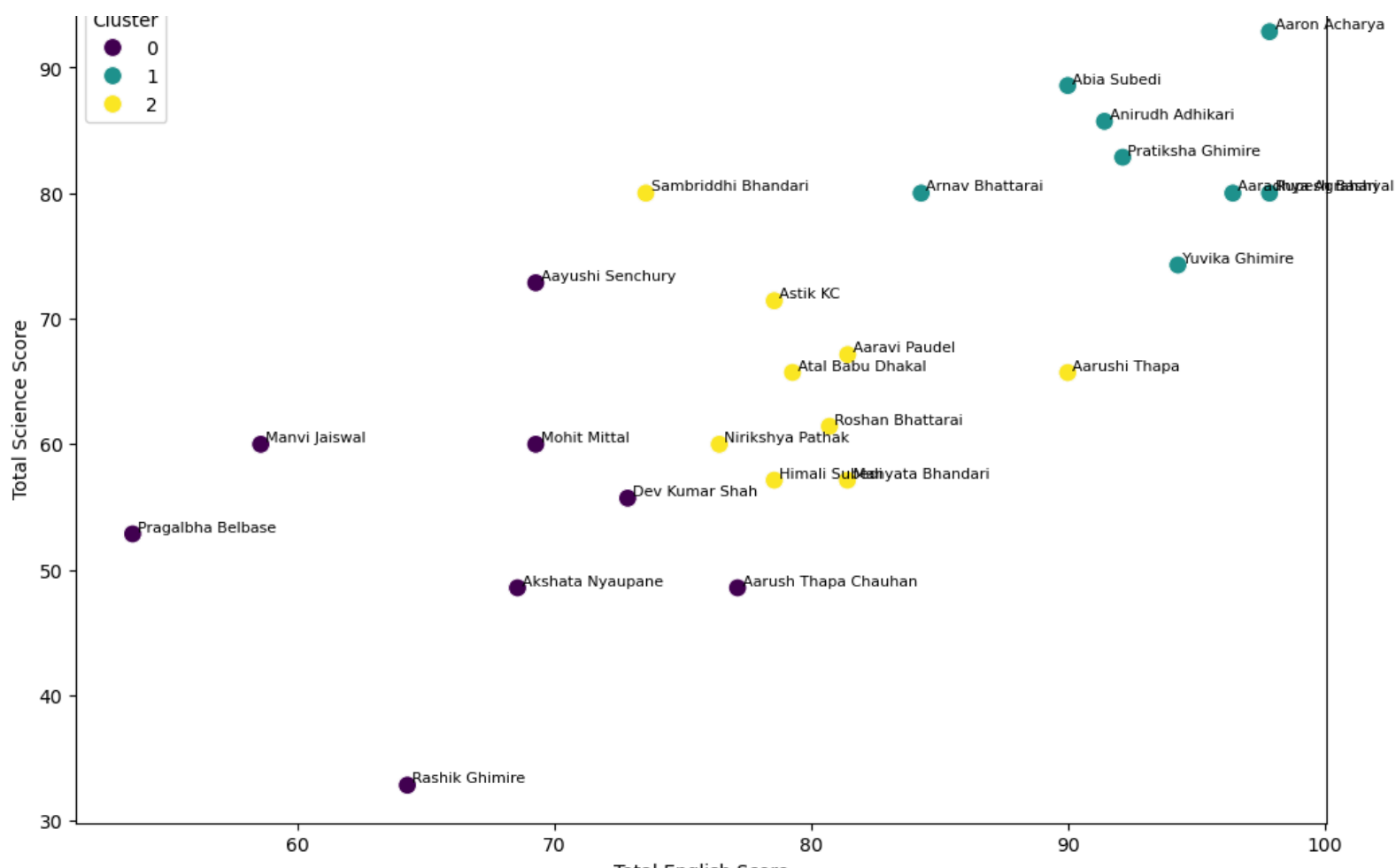
The above figure is based on the total marks of each subject obtain by the students which is categories in Average, Good, Excellent .. Thus, provides clear insight of distribution of students on basis of marks.

Conclusion:

For example: Let's see the first figure English which conclude In average category most of students obtain in rang form 68 - 78 marks . where as, In good category most of the students have obtained 88 marks in English . Same as in excellent category student have score 95-98 range of marks.

Subject clustering:

This section contains the overall data insights of students based on their performance in Total Science and Total English Score

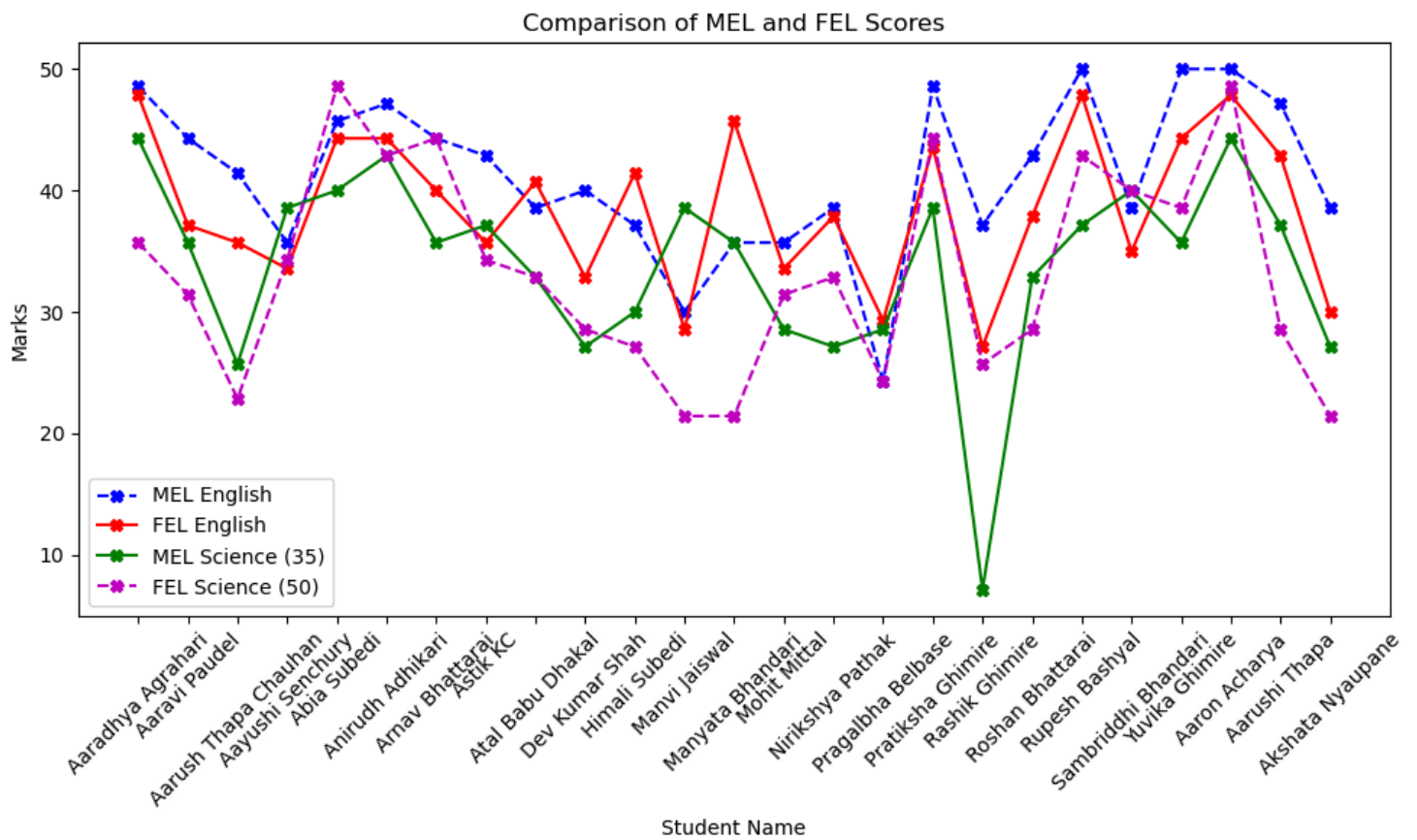


Conclusion:

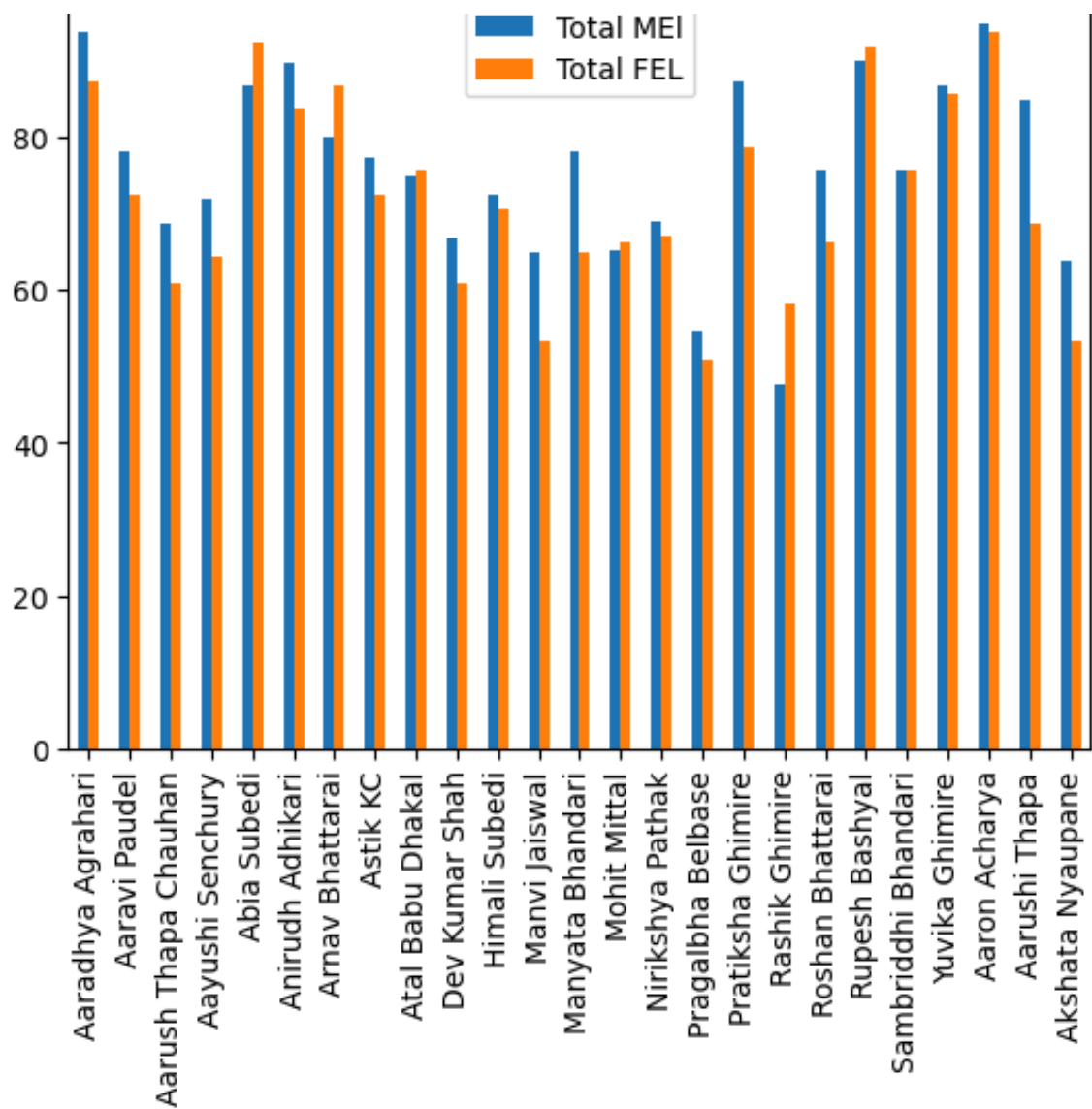
Above figure gives in-depth visual conclusion of individual students where it shows the marks obtained in English and Science. The dark purple dots represent the lower data of students which need improvements and individual focus. Whereas yellow dots data are the average student's data in both Science and English. Same as for gray dots represent the excellent student data.

MEL VS FEL ANALYSIS:

This report compares the performance of MEL and FEL based on their overall marks obtained in Midterm and final term Exam. The goal is to identify trends, gaps, and actionable insights to enhance learning outcomes.



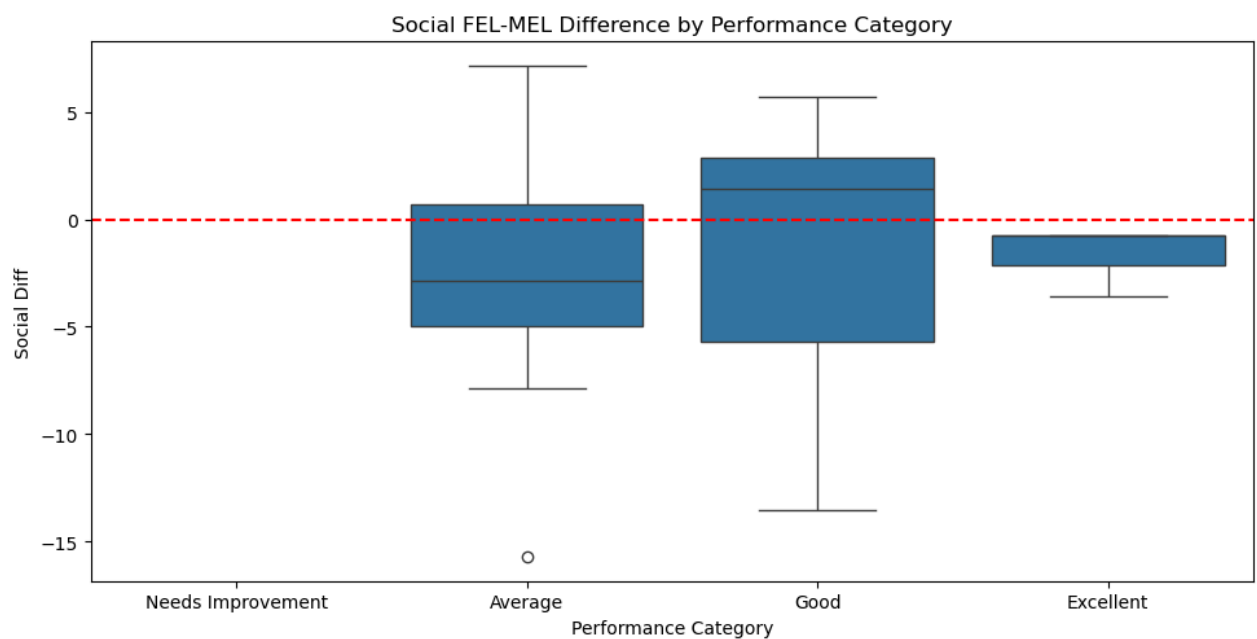
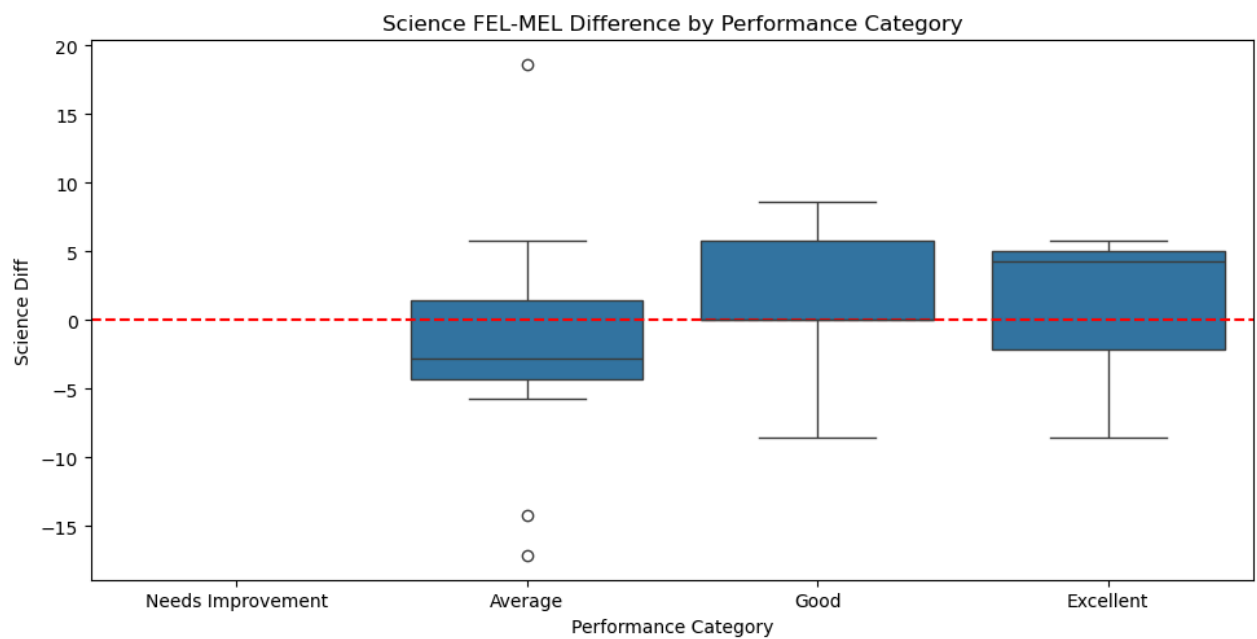
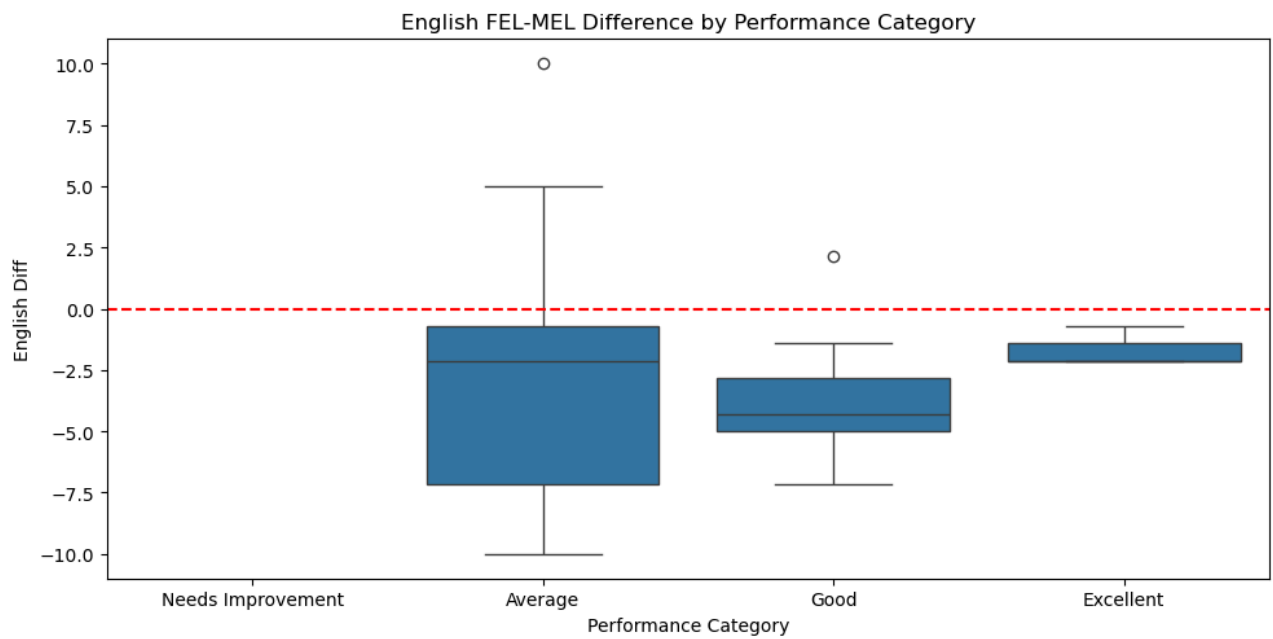
As above patterns shows clear data insights of individual students which will help to see the trending patterns of overall data set in comparison to MEL and FEL. For conclusion purpose we can compare the data patterns of students with MEL pattern to see the increase or decrease of the data. The above data of student's name isn't accurate so please look the data pattern one step back for example, The lowest pattern of red line MEL science is of Rashika Ghimire.



same visualization can be performed alongside comparing MEL and FEL for example Aaradhya Agrahari has slightly decrease on the FEL exam in comparison to MEL exam.

Variance changes in between MEL and FEL:

The below figure shows the changes of students marks i.e. increasing or decreasing changes between students according to the category of Average, Good, Excellent.

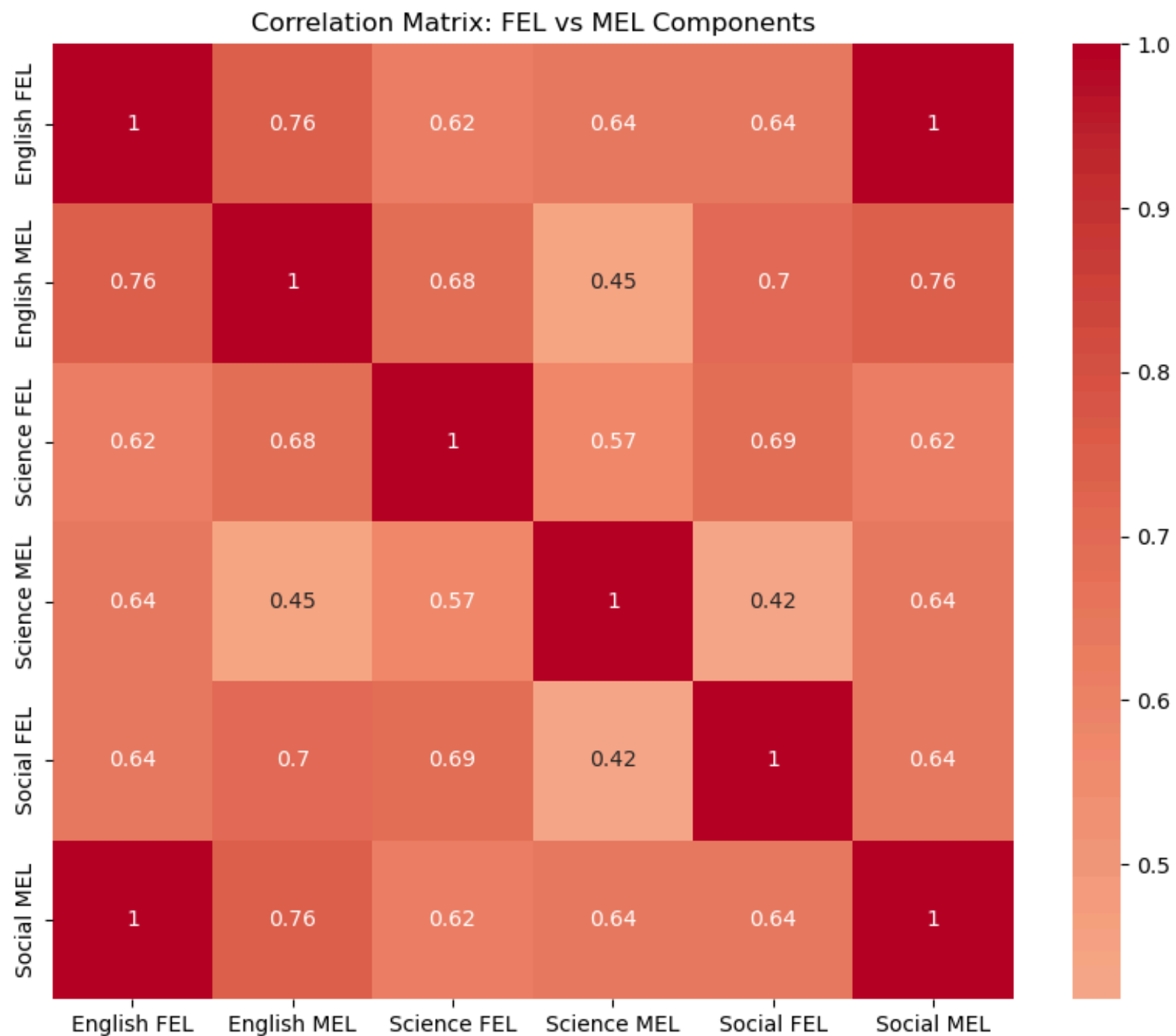


Conclusion:

The above figure shows the difference of student's data in MEL and FEL. for example, the first figure shows the data variance of English subject where the average, good, excellent data is presented were in Average group of data most of the students slightly decrease the marks in range of 2- 3 marks from MEL as shown inside the Whereas 2 or 3 students from average group has increase marks up to 5 marks

Correlation Between each subject:

Correlation refers to the how data is interlinked with each other i.e for example sports students may have higher IQ then others which states that those who are good at sports are also good at IQ. In same concept i have obtain a data visualization of student's whole data to bring meaning full analysis.



Conclusion:

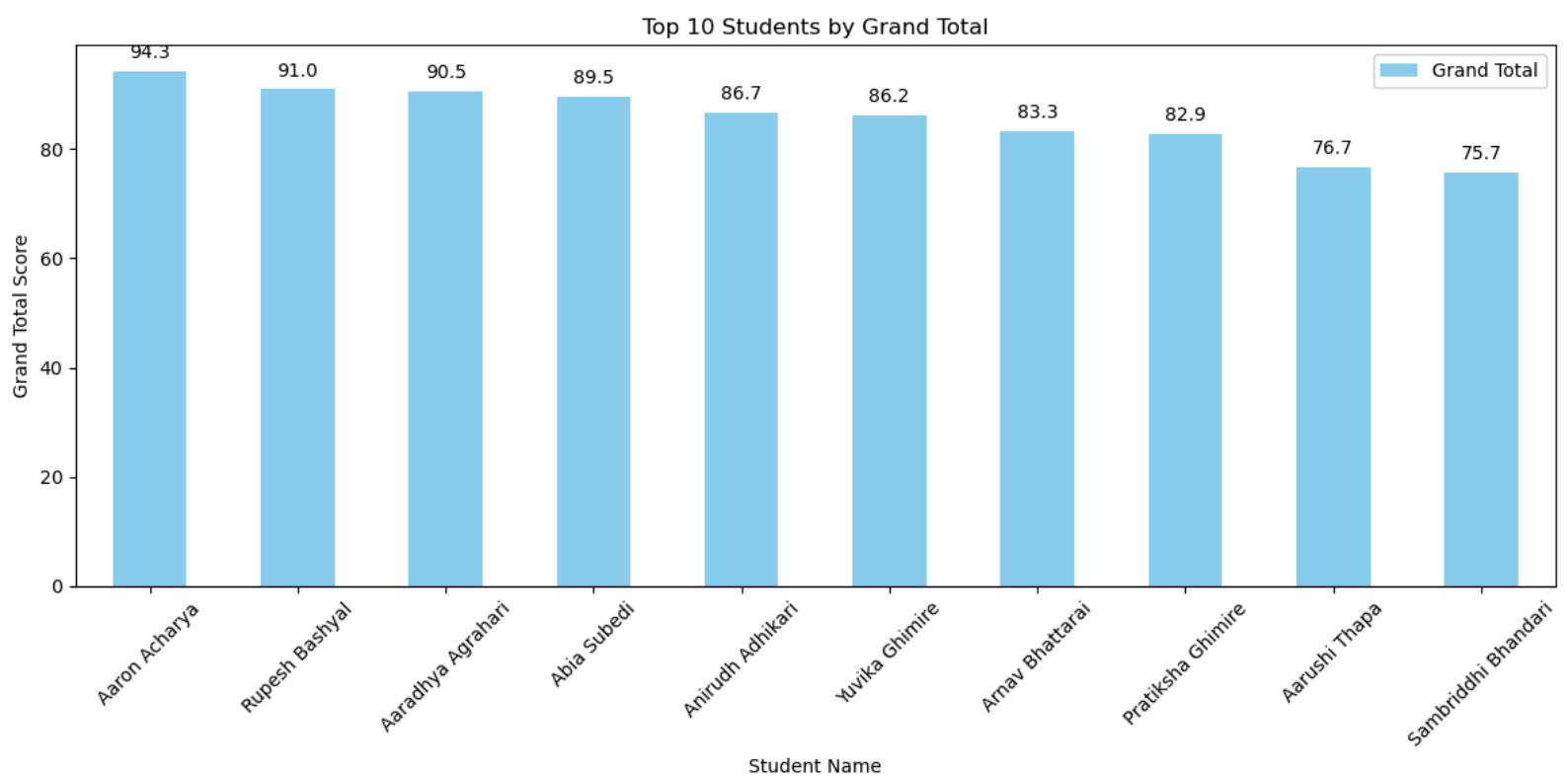
The above diagram shows the data of students in each term and each subject. These shows overall conclusion of those who are good at one subject is also good at another.

For reference:

Let's see the data of English FEL as first column data, In this data it strongly correlated with the social of MEL i.e those who have scores good in English FEL had also score good at Social MEL. whereas in another subject with 0.64 correlation is average whereas 0.76 is good and less then 0.5 is bad. As conclusion can be made accordingly.

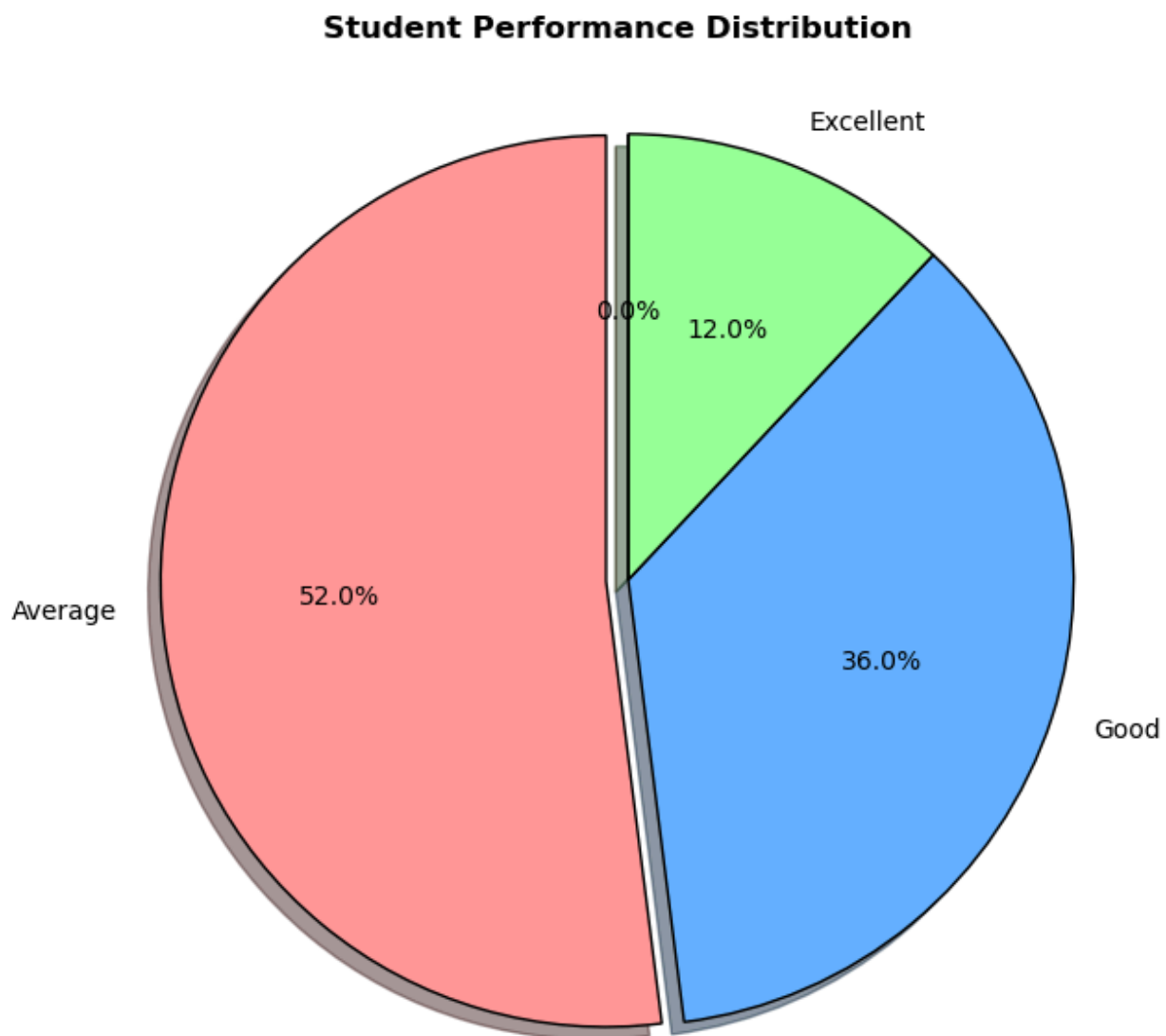
Grade Point Average Analysis

This content shows the Overall Students Grand Total Percentage Analysis were the final evaluation of data is presented and analyzed.



The top student achieved a Grand Total score of 94.3, followed by others in the 90s and 80s, indicating a competitive academic range.

Scores drop gradually from 94.3 to 75.7, with a noticeable gap between the top 3 students and the rest.



- **Excellent:** 36.0% of students achieved top-tier performance.
- **Good:** A significant portion of students (52.0%) performed well but fell short of the highest tier.
- **Average:** The remaining 12.0% (implied) represent students with moderate performance.

	Student Name	Grand Total	Predicted Total
0	Aaradhya Agrahari	90.48	91.011859
1	Aaravi Paudel	75.24	72.756946
2	Aarush Thapa Chauhan	64.76	66.513931
3	Aayushi Senchury	68.09	69.136130
4	Abia Subedi	89.52	84.316511
5	Anirudh Adhikari	86.67	85.546324
6	Arnav Bhattarai	83.33	76.642884
7	Astik KC	74.76	71.428883
8	Atal Babu Dhakal	75.24	76.382062
9	Dev Kumar Shah	63.81	63.256487
10	Himali Subedi	71.43	76.130527
11	Manvi Jaiswal	59.05	62.342532
12	Manyata Bhandari	71.42	84.401173
13	Mohit Mittal	65.71	64.836084
14	Nirikshya Pathak	68.10	70.050085
15	Pragalbha Belbase	52.86	59.020764
16	Pratiksha Ghimire	82.86	82.723326
17	Rashik Ghimire	52.85	46.884520
18	Roshan Bhattarai	70.96	72.509711

19	Rupesh Bashyal	90.95	87.937327
20	Sambriddhi Bhandari	75.71	71.694005
21	Yuvika Ghimire	86.19	82.471791
22	Aaron Acharya	94.29	91.011859
23	Aarushi Thapa	76.67	81.143729
24	Akshata Nyaupane	58.57	59.370549

Above data shows the predicted final outcomes of student through MEL Science and MEL Social subject only. Machine learning were used to predict the final grand percentage from mid term data.